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Revisit the VEC rule in high entropy alloys (HEAs) with high-throughput CALPHAD approach and its applications for material design-A case study with Al–Co–Cr–Fe–Ni system

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ABSTRACT

Valence electron concentration (VEC) was treated as a useful parameter to predict solid solution phases, and the VEC rule was proposed for high entropy alloys (HEAs). However, this empirical rule has its limitations, which restricts its applications for phase predictions in HEAs. In this paper, we revisited the empirical VEC rule with the HT-CALPHAD approach in the Al–Co–Cr–Fe–Ni system. Our investigation showed that more than 90% compositions are observed to have BCC structures when 5.7≤VEC≤7.2 and we got 100% FCC structures when VEC≥8.4. Meanwhile, we proposed a data screening procedure to classify and discover the new HEAs. The concepts of average density (ρ_{HEA}), highest BCC/FCC temperatures ($T^{max}_{BCC/FCC}$), and temperature ranges ($\Delta T_{BCC/FCC}$) were introduced as useful data screening criteria to down-screen the candidate alloy compositions for specific engineering applications. In the current work, we have identified three HEA categories (refractory BCC HEAs, light-weight BCC HEA and refractory FCC HEAs) in the Al–Co–Cr–Fe–Ni system and proposed the best candidate in each group. More importantly, our investigation showed the HEA non-equiatomic composition space provides ample opportunities for the discovery of the next-generation HEAs.

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1. Introduction

High entropy alloys (HEAs) are considered as promising metallic alloys used for structural and high-temperature applications due to their high toughness, high temperature softening resistance, high oxidation resistance and high hardness, etc. [1–3]. They are defined as alloys with five or more principal elements with concentrations between 5 and 35 at% [4,5]. HEAs typically adopt single-phase crystal-lattice structures such as face-centered-cubic (FCC) structure, body-centered-cubic (BCC) structure, or close-packed hexagonal (HCP) structures instead of complex intermetallic compounds [6]. Thus, every atom in the multi-principal-element matrix is surrounded by different kinds of atoms and suffers lattice strain and stress, which cause severe lattice distortion. This is widely accepted as the main reason for the unprecedented mechanical behaviors of HEAs.

Compared with traditional alloys, series of encouraging properties were reported for HEAs, which hold the potential in a wide range of applications for structural materials in harsh conditions. It has been reported that, in the NaCl solution, the CoCrFeNi single-phase face-centered cubic (FCC) alloy has markedly better pitting resistance than 304 L stainless steel, owing to its high Cr content [7]. Moreover, the addition of Cu to the CoCrFeNi alloy leads to the formation of Cu-rich phase and significantly reduces the pitting potential in NaCl solution. Ma et al. [8] observed that a strong creep resistance and a lower strain-rate sensitivity (SRS) value are acquired in $Al_{0.3}$CoCrFeNi HEAs. Zhang et al. [9] mentioned that the $Al_{0.5}$CoCrCuFeNi HEA exhibits high strength and plasticity greater than 30% in a temperature range of 300 K to 4.2 K. Lee et al. [10] found out that the NbTaTiV HEA shows the high yield strength and ductility at both room temperature and high temperatures (up to 900 °C). Atwani et al. [11] designed a novel W-based HEA with outstanding radiation resistance. No sign of irradiation-created dislocation loops was observed even after 8 displacement per-atom (dpa) in irradiative environment. Chung et al. [12] discovered that the hierarchical eutectic lamellar structure in the CoCrFeNiNb$_{0.5}$ HEA provides a promising solution for developing structural alloys with enhanced strength-toughness synergy.

The investigation of phase stabilities is crucial for the design of novel HEAs. Extensive effort has been carried out in this area in the last decade. Various simplified criteria [13] have been explored such as entropy of mixing ($\triangle S_{mix}$) [9], atomic size differences (δ) and melting points (T_m) [14], the enthalpy of mixing ($\triangle H_{mix}$) [15], the average number of itinerant electrons per atom ratio (e/a), parameter Ω (the scale ratio of $\triangle S_{mix}$ to $\triangle H_{mix}$) [16] and valence electron concentration (VEC)[17,18]. Out of all these parameters, the empirical VEC rule was first proposed by Guo et al. [17,18] based on their experimental observation of around 20

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multi-component systems such as $Al_xCrCu_yFe_zMnNi$, $Al_xCo_yCr_zCu_{0.5}Fe_vNi_w$, MoNbTaVW and CoCrFeMnNi, etc. It was claimed that FCC is stable at VEC≥8, BCC is stable at VEC≤6.87, and a mixture of FCC and BCC phases exists at 6.87≤VEC≤8. The effectiveness of the VEC rule for the prediction of FCC and BCC solid solution has quickly drawn the attention. Since the VEC rule was proposed, experiments have been carried out to test its reliability in $Al_{0.5}FeCrNiMnCoTa_x$, $FeCrNiMnCoTa_x$, MoNbHfZrTi and ReMoTaW systems etc. [19–23]. It has been concluded that the VEC rule does apply to these systems. In addition, the experiments from Tsai et al. [1,24]. showed that VEC can also be used to predict the formation of σ -phase. They observed that the VEC range of the formation of σ-phase was 6.88≤VEC≤7.84, which almost overlaps with the formation of the duplex FCC+BCC phases proposed by Guo et al. [25]. The empirical VEC rule has been adopted as one of the most frequently used rules for the design of HEAs with different crystal structures [13,26–30].

However, the empirical VEC rule has its own limitations which restricts its application for different multi-component systems: (1) Temperature plays an important role to determine the phase stabilities of FCC and BCC solution phases, but the VEC criterion didn't consider the effect of temperatures. Tian et al. [31] investigated the structural stability in $Al_xCoCrFeNi$ HEAs at 300 K based on the ab-initio approach and they found out that the FCC phase is stable at VEC≥7.57 and BCC phase is stable at VEC≤7.04. Although these threshold VEC value is not too far away from the VEC values proposed by Guo et al. [17,18], it does show that the threshold of VEC value can be changed with different temperatures; (2) The thresholds of VEC values are not consistent amongst different alloy systems [25,32,33]. Kube et al. [32] demonstrated that the VEC thresholds vary significantly depending on the types and concentrations of involved elements and the degree of lattice distortions so that they believe the threshold value may become a range rather than a fixed value; (3) The VEC criterion only works on the premise that solid solutions are the only alloys products, i.e., no intermetallic compounds or amorphous phase is formed [25]; however, in real case, the formation of intermetallic compounds and amorphous phases in HEAs is inevitable. It will significantly affect the mechanical properties of HEAs, which cannot be ignored; (4) The VEC rule is valid only based on HEAs containing mainly 3d or 4d transition metal elements[25]. For low-density multi-component HEAs(LDHEAs) containing high concentrations of Al, Mg, *Sc*, Li, Zn, Cu or Sn, it was found that the VEC rule fails to work effectively [16]; Youssef [34] observed FCC crystal structures in $Al_{20}Li_{20}Mg_{10}Sc_{20}Ti_{30}$ LDHEAs which VEC is equal to 2.8. However, based on Guo's criterion, it should have BCC structure. (5) The VEC criterion is summarized based on experiments from very limited amount of compositions and most of them are equiatomic or semi-equiatomic. The reliability of empirical VEC rule for non-equiatomic compositions are not fully understood.

Based on these limitations, the current work proposed a new way to revisit the VEC concept based on high-throughput CALculation of PHAse Diagram (HT-CALPHAD) approach. Al–Co–Cr–Fe–Ni quinary systems will be used as the case study to investigate the reliability VEC rule and its application to the material design.

2. Methodology

2.1. Generation of VEC-compositions

The calculation required physicochemical parameters for the constituent alloying elements which are listed in Table 1.

The generation of alloying compositions of Al–Co–Cr–Fe–Ni system is based on the VEC value of each element showing in Table 1, which can be calculated by Eq. (1) [25]:

$$VEC_{(HEA)} = \sum_{i=1}^{n} C_i(VEC)_i \quad (1)$$

where C_i and $(VEC)_i$ are the atomic fraction and the VEC of element *i* respectively. In this study, the VEC step length is set as 0.1 to consider 71 different VEC values and the composition step length for each element is set as 0.05. Overall, 3561 compositions that match Eq. (1) are randomly generated, which are called as *VEC-compositions*.

Table 1
Physiochemical properties of each element in Al–Co–Cr–Ni–Fe HEAs.

Element	Density(g/cm^3)	VEC
Al	2.70	3
Co	8.83	9
Cr	7.19	6
Fe	7.87	8
Ni	8.90	10

2.2. HT-CALPHAD modeling

2.2.1. Computational thermodynamic approach

The CALPHAD approach was pioneered by Larry Kaufman [35], which aims to promote computational thermodynamics through the development of models to represent thermodynamic properties for various phases. In this work, Thermo-Calc and TC-python software [36] will be used for HT-CALPHAD calculation. Thermo-Calc is a program developed by KTH back from the 1970s, which has been at the forefront of scientific software and databases for calculation of computational thermodynamics and phase diagram. TC-Python is a Python™ language-based SDK available with Thermo-Calc giving Thermo-Calc users access to many other programs, such as numerical packages, like *NumPy* and *SciPy*, which can be used to proceed the HT calculations. The TCM commands (Thermo-calc console module input command) can be converted to TC-Python codes, allowing a user to run batch calculations for many varied parameters.

2.2.2. High-throughput calculation of equilibrium solidification (HT-CES)

Based on the available TC-Python API, the calculation of equilibrium solidification of each generated VEC-composition is performed. The calculation details are provided in the supplementary method section (see S1 in the supplementary file).

2.3. Data screening for materials design

After HT-CES calculations, the data screening was carried out by the in-house *Python* codes based on the needs of engineering applications. It was found that the weights of BCC HEAs dramatically decrease with the addition of Al, as Al is considered as a BCC stabilizer in the Al–Cr–Co–Fe–Ni system [37]. Meanwhile, BCC structures typically come with high melting temperatures. Two screening categories were thus defined: refractory BCC HEAs and light-weight BCC HEAs. However, only one screening category was defined for FCC HEAs, which is due to FCC structures typically come with high VEC values which are corresponding low concentration of Al. The details of data screening are provided in the supplementary method section (see S2 in the supplementary file).

3. Case study of Al–Co–Cr–Fe–Ni system

3.1. Background

Al–Co–Cr–Fe–Ni system has evoked researchers' interest in the last decade. People have investigated the semi-equiatomic compositions (i.e., $Al_xCoCrFeNi$ and Al_xCoCr_yNiFe) in which only one or two elements are free to change, and the others are equiatomic. For the $Al_xCoCrFeNi$ system, it has been observed that the crystal structures and mechanical properties of the HEAs in this system could vary dramatically with x, the Al content. Many reports have reviewed the

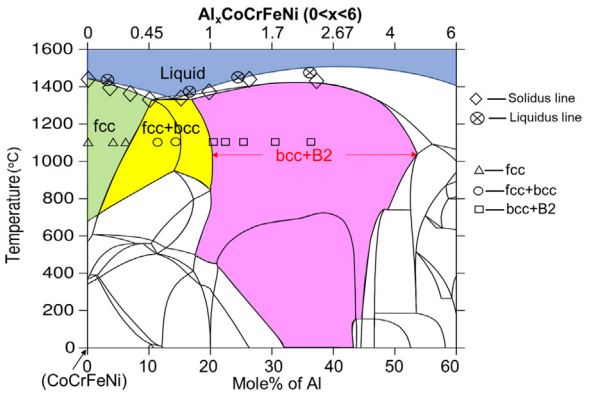


Fig. 1. The CoCrFeNi–Al(60 mol%) pseudo-binary phase diagram comparison with experimental data [25,37].

phase transformation in this system, where x varies from 0 to 2 while Cr, Co, Ni, and Fe are equiatomic [38–41]. Kao et al. investigated the structures of as-cast, homogenized, and deformed $Al_xCoCrFeNi$ alloys [37]. It shows the crystal structure sequentially transforms from a single FCC phase to duplex FCC+BCC phases and finally to a single BCC phase with x increases, which implied that Al is a BCC stabilizer in $Al_xCoCrFeNi$ alloys. At the same time, the hardness increases dramatically with the addition of Al, which is due to the transformation of FCC to BCC structures. Shu et al. found FCC is the only stable phase in the as-cast structure of the $Al_{0.3}CoCrFeNi$ [42]. However, the B2 phase precipitates out after aged at 900 °C for 72 h. Chen et al. also observed that $Al_{0.7}Co_{0.3}CrFeNi$ consists of two BCC phases, which have the same lattice parameter and one FCC phase from TEM analysis [43]. Recently, Wang systematically studied the phase stability of the $Al_xCoCrFeNi$ systems [44]. Only a single FCC phase was found with x = 0.1. In addition, an FCC+B2 duplex structure is observed with x larger than 0.75. Two BCC phases, including the precipitates with A2 structure and matrix with B2 structure, are found with x = 1.5. Similarly, the effect of both Al and Cr on the phase stabilities of Al_xCoCr_yFeNi (x = 1, 1.5, 2, 2.5, and 3, y = 1 and 2) has been investigated as well [45]. It showed that both Al and Cr are stabilizers of the BCC-based structures, and a higher Cr ratio stabilizes the disordered BCC phase against both ordered B2 and FCC phases.

While there are significant achievements from the semi-equiatomic Al–Co–Cr–Fe–Ni HEAs, the phase stabilities outside the equiatomic composition attract much more interest. The challenge is there are tons of compositions that need to be investigated for non-equiatomic Al–Co–Cr–Fe–Ni HEAs, and it is impossible to follow the traditional trial-and-error approach. Fortunately, the HT-CALPHAD approach can perform thousands and even ten thousand calculations in user-defined compositions in a specific system, which greatly reduces the time for the design of next-generation HEAs, but the validation of CALPHAD database is needed.

3.2. Validation of TCHEA3 database in Al–Co–Cr–Fe–Ni system

There are several commercial HEA thermodynamic databases available, such as TCHEA from Thermo-Calc and PANHEA from Pandat. However, the verification of the thermodynamic database reliability with experimental data is imperative before making any predictions. In the current work, the isopleth CoCrFeNi–Al pseudo-binary phase diagram has been generated with the TCHEA3 database, as shown in Fig. 1. It can be seen the primary solid phase is FCC when x<0.45 around 1200–1400 °C, while the BCC+B2 phase will be developed when x>1 and a mixture of FCC and BCC phase should form in between when 0.45<x<1 around 1000–1200 °C. This is what exactly has been observed by Kao et al. [37] in $Al_xCoCrFeNi$ at 1100 °C. Also, the simulation results also match very well with the previous experimental observation by DSC and DTA on the liquidus and solidus temperatures [25], which confirmed the reliability of the TCHEA3 database for the prediction of $Al_xCoCrFeNi$ HEAs.

In addition, two pseudo-ternary isothermal phase diagrams, i.e., FeCrCo–Ni–Al and FeCrNi–Co–Al at 1000 °C, were plotted in comparison with the simulation results from PANHEA database [46] and experimental data [43,47] for $Al_{1.7}CoCrFeNi_{0.3}$, $Al_{0.3}CoCrFeNi_{1.7}$ and $Al_{0.7}Co_{0.3}CrFeNi$ HEAs, as shown in Fig. 2. The comparison shows that the simulation result from TCHEA3 has good agreement with all the results from Pandat and most of the results from experimental data, except $Al_{1.7}CoCrFeNi_{0.3}$. It indicates that the Gibbs energy description of BCC and FCC phases in the TCHEA3 database for this quinary system is reliable for further predictions.

3.3. Manual verification of empirical VEC rule

The verification of the empirical VEC rule should be carried out before conducting HT-CES procedure. In this section, 41 compositions (1 composition for each VEC value) from 5≤VEC≤9 (step=0.1) were randomly generated and manually calculated with Thermo-Calc. The phase transformation points of these compositions were recorded, which is shown in Fig. 3(a). The red, blue, and orange symbols represent the highest temperature (solidus temperature) of single BCC, FCC+BCC, and single FCC respectively. The purple symbols labeled as *MUL* represent the temperatures that secondary phases will show up, including solution phases other than FCC and BCC and stoichiometric compounds.

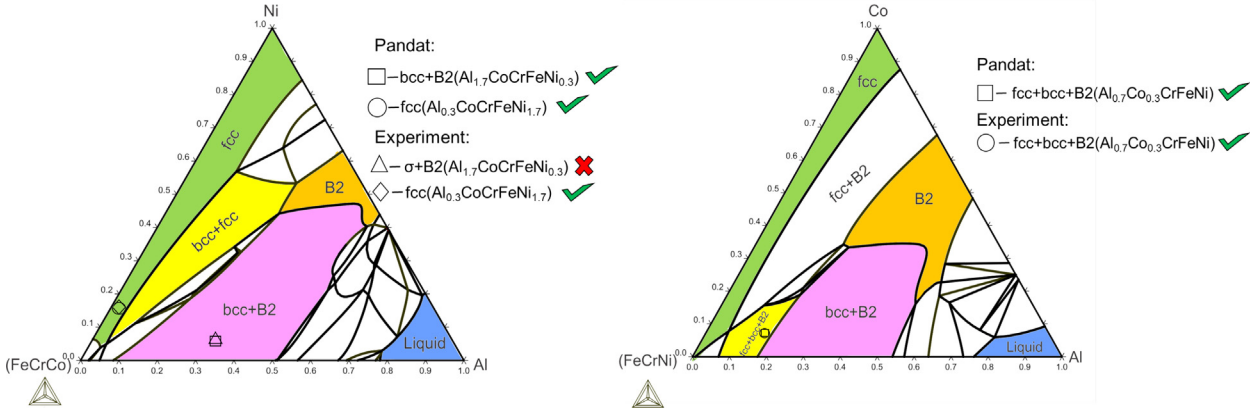


Fig. 2. Pseudo-ternary isothermal phase diagram at 1000 °C comparison with Pandat [46] and experimental data [43,47] (a) FeCrCo–Ni–Al (b) FeCrNi–Co–Al.

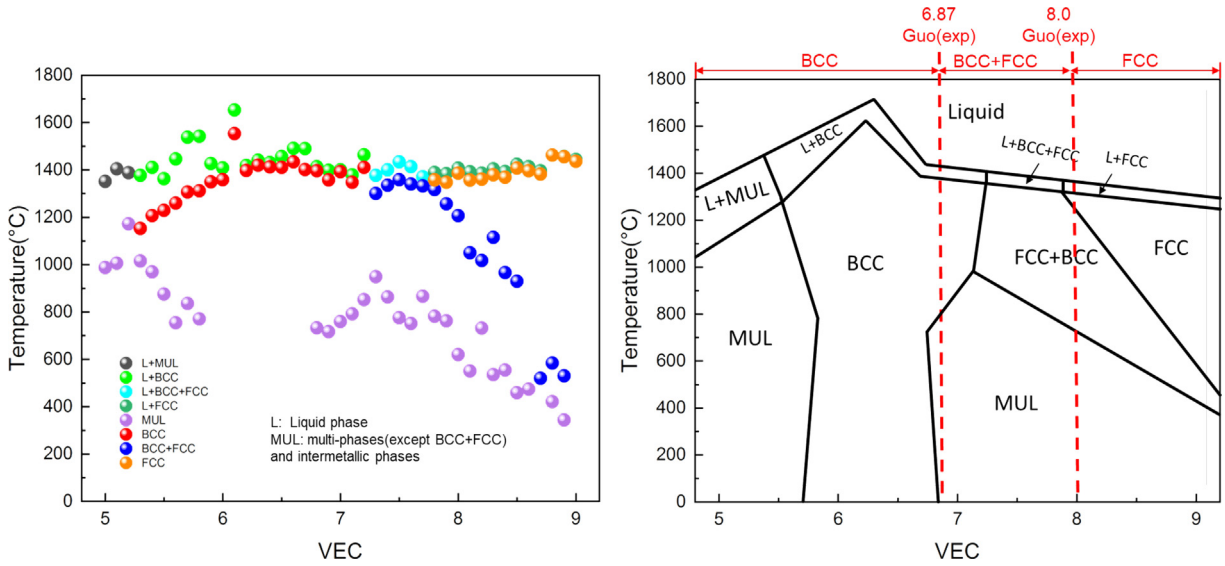


Fig. 3. VEC-Temperature diagrams of Al–Co–Cr–Fe–Ni systems: (a) Phase transformation points (b)VEC-Temperature diagram.

It is worth noting that both $L1_2$ (ordered FCC) and A1 (disordered FCC) are treated as FCC, while B2 (ordered BCC) and A2 (disorder BCC) are treated as BCC, respectively in the current work as FCC and $L1_2$ as FCC_$L1_2$, and BCC and B2 as BCC_B2 in the TCHEA3 database. In addition, it is believed that coherent secondary phases such as $L1_2$ or B2 phase will have precipitation hardening effect, which can strengthen the FCC or BCC matrix.

Based on the phase transformation points calculated by Thermo-Calc, the VEC-temperature phase diagram was plotted as shown in Fig. 3(b). It should be noted that this diagram is based on the randomly generated VEC-compositions, which is not a regular phase diagram as each VEC value could have very different composition combinations. It is interesting to observe the BCC phases are stable at around $5.4 \leq VEC \leq 7.2$; BCC+FCC phases coexist at $6.7 \leq VEC \leq 8$; a single FCC phase exists at $VEC \geq 8$. Comparing with Guo's empirical VEC rule (Fig. 3(b) red dash lines), it can be observed that the current simulation results have a good agreement, which confirms that the reliability of Guo's VEC rule in this system.

In addition, it can be observed from Fig. 3(b): 1.) Most of BCC structures are stable down to room temperature, while FCC and FCC +BCC are only stable at high temperatures as the secondary phases will precipitate out at low temperatures. 2.) The melting temperatures of BCC structures are relatively higher than FCC and FCC+BCC structures, which means BCC structures can be good candidates for high-temperature applications. Overall, the Guo's VEC rule was confirmed manually with the CALPHAD approach.

3.4. High-throughput simulations

In the current work, the compositions were generated based on the VEC value (see Section 2.1), and their phase stabilities were simulated by TC-Python. Compositions containing temperature ranges during which FCC or BCC is stable without other phases were screened out with the screening *Step 1* (see S2 in the supplementary file), which is defined as $\Delta T_{BCC/FCC}$. In this work, 1607 and 1262 out of the total 3561 compositions have ΔT_{BCC} (Fig. 4(a)) and ΔT_{FCC} (Fig. 4(b)) respectively. The red dots and blue dots indicate the highest temperatures and lowest temperatures of which single FCC (Fig. 4(a)) and BCC exist (Fig. 4(b)), respectively. It is defined as $T^{max}_{BCC/FCC}$ and $T^{min}_{BCC/FCC}$.

Because there are many different compositions with the same VEC value, the concept of the average density of each composition (ρ_{HEA}) was introduced as the third variable(the 1st and 2nd variables are VEC and temperature respectively) which can be calculated by Eq. (2):

$$\rho_{HEA} = \sum_{i=1}^{n} C_i \rho_i \tag{2}$$

where C_i is the atomic fraction of element i and ρ_i is the density of element i. It is worth noting that the composition can be also separated based on some other parameters such as the cost and the recyclability of the elements.

Because of the limitation of temperature ranges selected in HT-CES, the maxim solidus temperatures of some compositions reach 1723 °C, but the real solidus temperature could be even higher. Meanwhile, a lot of screened-out BCC or FCC structures are stable at 0 °C (blue dots), indicating that it is possible to have some compositions at which pure FCC or BCC structures are stable in the whole temperature range.

Fig. 5 shows the possibility (p) that single BCC and FCC phase exists for each VEC value, which can be calculated by Eq. (3):

$$p = \frac{N_{(BCC/FCC)}}{N_{(total)}} \tag{3}$$

where $N_{(BCC/FCC)}$ is the number of generated compositions that present the single BCC/FCC phase in a specific VEC value and $N_{(total)}$ is the total number of compositions generated in that specific VEC value. Based on the statistic results, more than 90% of the generated compositions have single BCC structures at $5.7 \leq VEC \leq 7.2$, and this possibility starts to decrease when $VEC \leq 5.7$ or $VEC \geq 7.2$. Meanwhile, the possibility of getting the single FCC structure starts to increase when $VEC \geq 7.5$, and it reaches 100% at $VEC \geq 8.4$. The results from 3D-mapping in Fig. 4 show consistency with the previous 2D-mapping in Fig. 3, indicating that the VEC rule is valid statistically in Al–Co–Cr–Fe–Ni system.

It should be noted that Al has very limited solubility range with Cr, Co, Ni and Fe, and the HT-CES composition step for each element is 0.05. Hence, the FCC temperature range at low VEC values (Al-rich side) cannot be observed in Fig. 4(b). The highest temperatures ($T^{max}_{BCC/FCC}$) and the temperature ranges of BCC and FCC structures ($\Delta T_{BCC/FCC}$) calculated from the HT-CES procedure are shown in Figs. 6 and 7. The results indicate that higher T^{max}_{BCC} (around

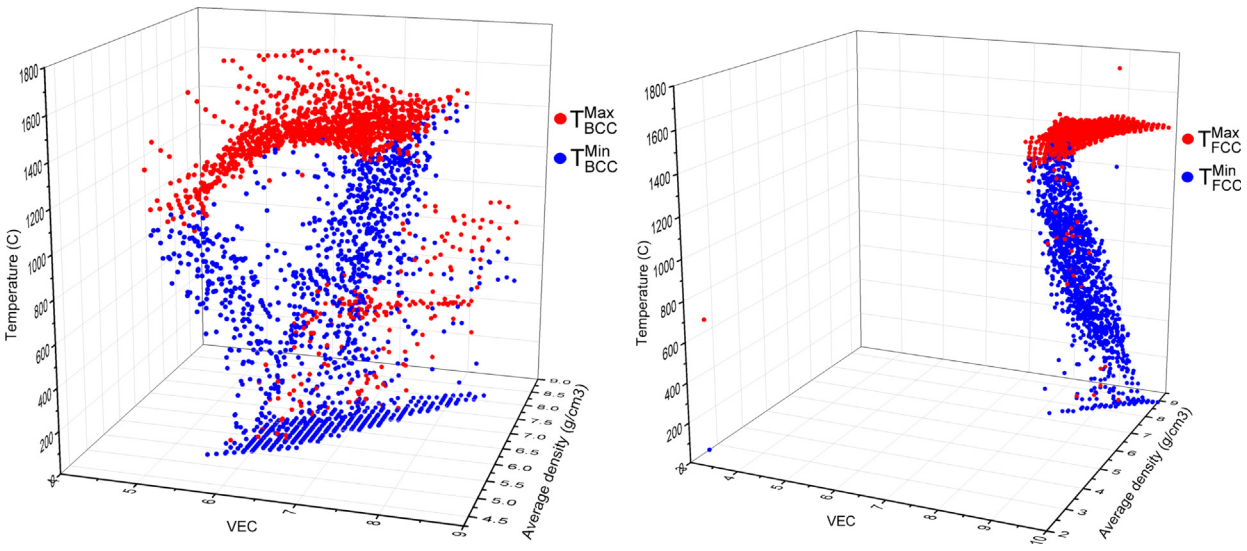


Fig. 4. Phase stabilities of FCC and BCC structures in Al–Co–Cr–Fe–Ni HEAs: (a) BCC structures, (b) FCC structures.

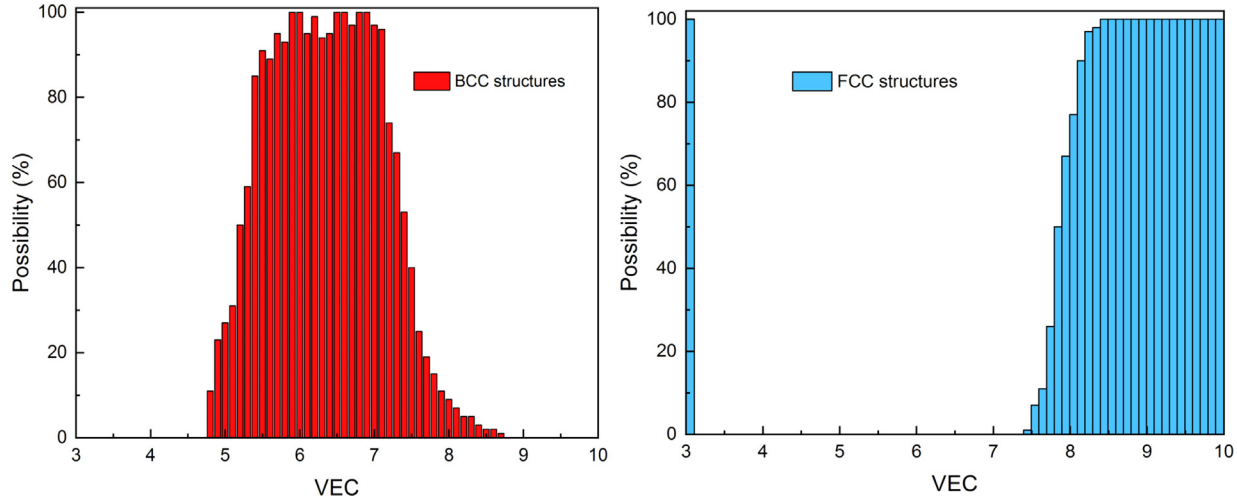


Fig. 5. The possibility that pure BCC and FCC phase phases exist for each VEC value: (a) BCC structures, (b) FCC structures.

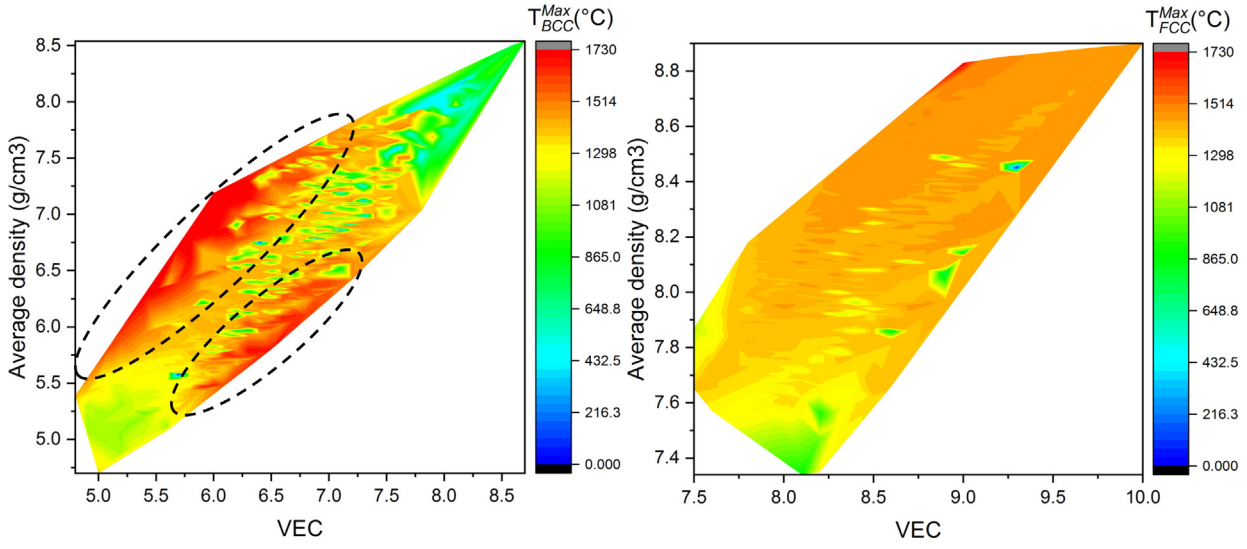


Fig. 6. VEC-Average density-Highest BCC/FCC temperature diagram: (a) BCC structures, (b) FCC structures.

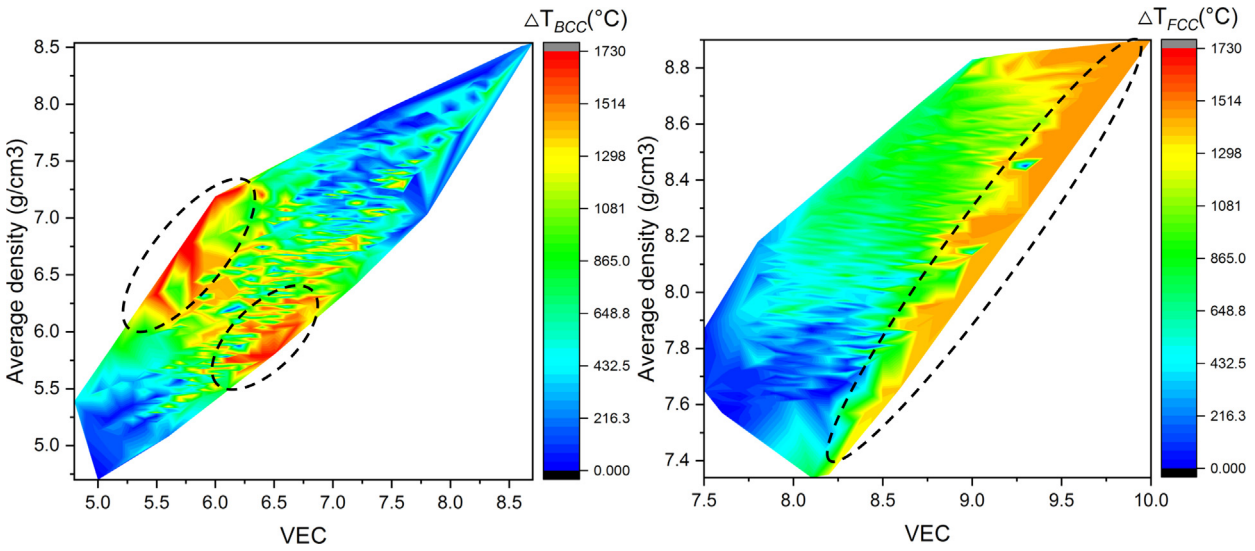


Fig. 7. VEC-Average density-△T diagram: (a) BCC structures, (b) FCC structures.

1600–1723 °C) are present when the VEC is between 6.0–7.0 at low-density region and 5.0–7.0 at high-density region (Fig. 6(a) black dash circle). However, for the FCC phase, most of them show higher temperatures at VEC≥8.3, which is around 1300–1400 °C (Fig. 6(b)). In addition, when the VEC is around 5.8–7.0, larger ΔT_{BCC} (Fig. 7(a) black dash circle) are observed, especially when the VEC is around 6.5 at low-density region (around 6.0 g/cm^3) and VEC around 5.8 at the high-density region (6.3–7.2 g/cm^3), ΔT_{BCC} becomes even larger (larger than 1600 °C), which means there are possibly no any other phase present except BCC solid structure. Meanwhile, with the increase of the VEC value, larger ΔT_{FCC} is observed at both high-density region and low-density region (Fig. 7(b) black dash circle). A better understanding of the impact of VEC value, average density and temperatures on phase stabilities of BCC and FCC structures can be achieved via the 2D contour diagram. Meanwhile, the highest temperature, average density and the temperature range of BCC and FCC structures are important parameters for industrial applications.

3.5. Data screening for engineering applications

Based on data screening procedures (see S2 in the supplementary file), 7 candidates for refractory BCC HEAs (Table 2), 16 candidates for light-weight BCC HEAs (Table 3) and 3 candidates for refractory FCC HEAs (Table 4) are screened out from this work. The VEC values, compositions, average density, and lowest/highest temperatures of each case are given in Tables 2–4:

From Table 2, it can be found out that $Al_{0.05}Co_{0.05}Cr_{0.75}Fe_{0.1}Ni_{0.05}$ (red marked) has the highest T_{BCC}^{max} (1630 °C). Meanwhile, the lowest BCC temperature (T_{BCC}^{min}) is 0 °C, indicating that BCC structures are stable at room temperature. Although this composition shows good refractory performance, the concentration of Al is very low, and the average HEA density is relatively high. The other candidates from Table 2 show lower T_{BCC}^{max} (around 1500 °C) and ΔT_{BCC}, but the concentration of Al is higher, and the average density is much lower compared with $Al_{0.05}Co_{0.1}Cr_{0.75}Fe_{0.1}Ni_{0.05}$ HEAs. Similar results can be found in Table 3 as well. $Al_{0.45}Co_{0.15}Cr_{0.1}Fe_{0.25}Ni_{0.05}$ (red marked) shows the lowest average density compared with other candidates, but the T_{BCC}^{max} is much lower. Accordingly, it's hard to get the compositions with both high T_{BCC}^{max} and lower average density, and there is a trade-off between the selection of average density and the T_{BCC}^{max}. For FCC candidates (Table 4), there are not much differences about T_{FCC}^{max} and ΔT_{FCC}, so in this work, the composition (red marked) which shows the lowest average density would be selected for engineering application.

The precipitation hardening effect can greatly increase the mechanical properties of HEAs, but the HT-CES calculations did not provide such information because both B2 and $L1_2$ structures have been treated as BCC structures and FCC structures in the TCHEA3 database respectively. Hence, the mole fraction of each individual phase at different temperatures was calculated in this section, which provides the detail thermodynamic equilibria information for the final candidate alloys.

The equilibrium solidification simulations of the three final candidates from Tables 2–4 are selected as examples to confirm the accuracy of our HT calculations, which are shown in Fig. 8. As shown in Fig. 8(a), it can be observed that there are two different BCC structures in the best refractory BCC HEAs, i.e. $Al_{0.05}Co_{0.1}Cr_{0.75}Fe_{0.1}Ni_{0.05}$: 1.) the disordered BCC structures (labeled as A2 in the diagram) and 2.) ordered BCC structures (labeled as B2 in the diagram). During the solidification process, A2 structure is the first solid phase forming from liquid, while all the liquid phase is transformed into A2 structure at 1630 °C. The first B2 structure (B2#1) starts to precipitate out from A2 structures at around 1200 °C, while the second B2 structure

Table 2
Final candidates for refractory BCC HEAs.

VEC	x(Al)	x(Co)	x(Cr)	x(Fe)	x(Ni)	Average density(g/cm^3)	Lowest T (°C)	Highest T (°C)
6.4	**0.05**	**0.05**	**0.75**	**0.1**	**0.05**	**7.20**	**0**	**1630**
6.4	0.4	0.4	0.05	0.1	0.05	6.20	0	1520
6.5	0.4	0.3	0.05	0.1	0.15	6.21	0	1530
6.5	0.4	0.4	0.05	0.05	0.1	6.26	0	1550
6.6	0.4	0.3	0.05	0.05	0.2	6.26	0	1540
6.7	0.4	0.2	0.05	0.05	0.3	6.27	0	1530
6.8	0.35	0.35	0.05	0.1	0.15	6.52	0	1510

Table 3
Final candidates for light-weight BCC HEAs.

VEC	x(Al)	x(Co)	x(Cr)	x(Fe)	x(Ni)	Average density(g/cm^3)	Lowest T (°C)	Highest T (°C)
5.8	0.4	0.2	0.25	0.1	0.05	5.88	0	1360
5.8	**0.45**	**0.15**	**0.1**	**0.25**	**0.05**	**5.67**	**0**	**1320**
5.9	0.4	0.1	0.25	0.1	0.15	5.88	0	1380
5.9	0.4	0.2	0.2	0.15	0.05	5.91	0	1360
5.9	0.4	0.1	0.2	0.2	0.1	5.87	0	1350
5.9	0.4	0.1	0.15	0.3	0.05	5.85	0	1330
5.9	0.45	0.25	0.1	0.15	0.05	5.77	0	1380
5.9	0.45	0.15	0.1	0.2	0.1	5.72	0	1360
6.0	0.4	0.1	0.15	0.25	0.1	5.90	0	1360
6.0	0.45	0.35	0.1	0.05	0.05	5.86	0	1460
6.0	0.45	0.25	0.1	0.1	0.1	5.82	0	1420
6.0	0.45	0.25	0.05	0.2	0.05	5.80	0	1400
6.1	0.45	0.25	0.1	0.05	0.15	5.87	0	1460
6.1	0.45	0.15	0.05	0.2	0.15	5.81	0	1440
6.2	0.4	0.1	0.15	0.15	0.2	6.00	0	1390
6.5	0.45	0.05	0.05	0.05	0.4	5.97	0	1490

Table 4
Final candidates for refractory FCC HEAs.

VEC	x(Al)	x(Co)	x(Cr)	x(Fe)	x(Ni)	Average density(g/cm^3)	Lowest T (°C)	Highest T (°C)
8.9	**0.1**	**0.1**	**0.05**	**0.05**	**0.7**	**8.14**	**0**	**1430**
9.2	0.05	0.15	0.05	0.05	0.7	8.44	86	1450
9.2	0.05	0.05	0.05	0.1	0.75	8.40	0	1430

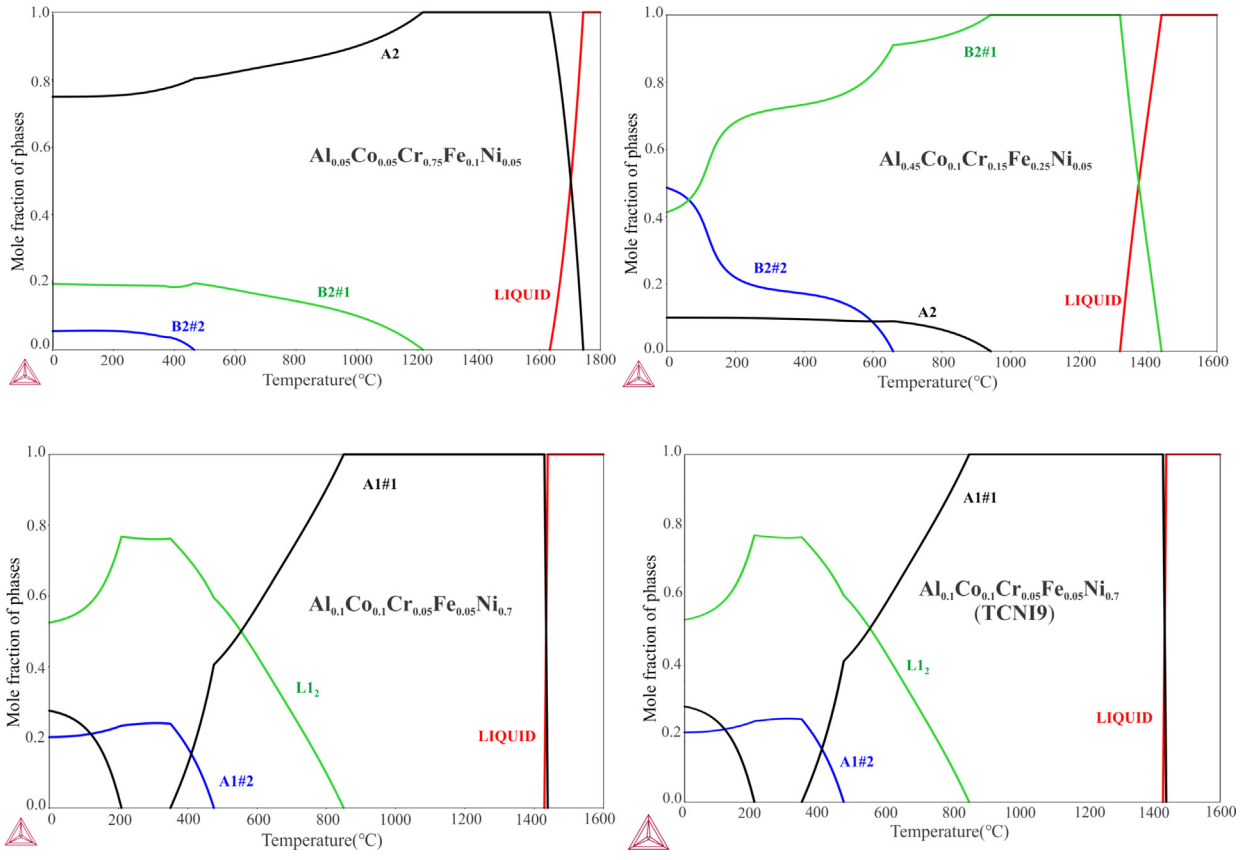


Fig. 8. Equilibrium solidification diagrams of final candidates (a) Refractory BCC HEA, (b) Light-weight BCC HEA, (c) Refractory FCC HEA, (d) Refractory FCC HEA calculated by TCNI9 database.

(B2#2) begins to precipitate out around 480 °C. It is worth noting there is a large temperature range from 1200 °C to 1600 °C that A2 structure is the only stable phase. From the engineering point of view, this can be a good candidate for heat treatable alloys. It is possible to do the heat treatment and control of the size of the precipitates with solutionization, quenching, or aging, etc.

In comparison, the solidification simulation for the best light-weight BCC HEA, i.e. $Al_{0.45}Co_{0.15}Cr_{0.1}Fe_{0.25}Ni_{0.05}$, is shown in Fig. 8(b). It can be observed that liquid phase is fully transformed to B2#1 phase at 1320 °C, and A2 and B2#2 phases precipitate out at lower temperatures. For the engineering applications, BCC structure HEAs have relatively lower ductility but higher strength and hardness. Such a combination of A2 + B2 phases may dramatically increase its yield strength and high-temperature properties. For example, Ma et al. reported the compressive yield strength could be enhanced drastically, and keeps constant at a high strength level of 1350−1400 MPa, when the A2 + B2 are dominant in Al_xCoCrFeNi HEAs with $x>0.74$ [48]. It is believed that the precipitation of strengthening B2 increases the mechanical property of the A2 HEAs.

For FCC structure HEAs (Fig. 8(c)), it can be seen that the liquid phase is completely transformed to A1 structures(disordered FCC structures) at 1430 °C, and $L1_2$ structure(ordered FCC structure) and another A1 structure will precipitate out at lower temperatures, which can be considered as strengthening phases as well. Compared with BCC structure HEAs, FCC HEAs are more ductile, but the strength and the melting point are not very high. However, in this case, the highest FCC temperature is 1430 °C, and it is high enough. Hence, $Al_{0.1}Co_{0.1}Cr_{0.05}Fe_{0.05}Ni_{0.7}$ HEAs is also a good candidate for further investigation. Because this candidate is considered as Ni-rich HEA, the composition is also testified with the TCNI9 database which is shown as Fig. 8(d), which provides the same results from the TCHEA3 database.

4. Conclusions

In the current work, we systematically investigated the phase stability, especially the FCC and BCC structures, in the Al−Co−Cr−Fe−Ni system with the HT-CALPHAD approach. The original empirical VEC rule proposed by Guo et al. [17,18] was carefully examined with manually generated VEC-compositions, the VEC-temperature phase diagram, and also the HT-CES simulations. Our calculations confirmed the limitations of the empirical VEC rule. Meanwhile, it does show the VEC itself has strong physical meanings. We proposed a new way to use VEC, i.e., how to consider the impact of temperatures and all the possible forming phases other than solution phases, which can help to expand the design HEAs to the non-equiatomic compositions in multicomponent systems.

Also, we proposed a data screening procedure to classify and discover the new HEAs, considering ρ_{HEA}, $T^{max}_{BCC/FCC}$, and $\Delta T_{BCC/FCC}$. Three HEA categories were identified in the Al−Co−Cr−Fe−Ni system, and the best candidate in each group was proposed. Out of all the HEA candidates, $Al_{0.05}Co_{0.1}Cr_{0.75}Fe_{0.1}Ni_{0.05}$ was identified as the best refractory BCC HEA candidate, while $Al_{0.45}Co_{0.15}Cr_{0.1}Fe_{0.25}Ni_{0.05}$ as the best light-weight BCC HEA candidate, and $Al_{0.1}Co_{0.1}Cr_{0.05}Fe_{0.05}Ni_{0.7}$ as the best refractory FCC HEA candidate.

In the future, the elemental cost, process cost, and mechanical properties could be used as additional screening parameters for alloy design as well, which will pave the way to identify the next generation HEAs quickly.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Author contributions

S.Y. constructed the idea of the HT calculations and the applications to material design based on the data screening method and had primary writing responsibilities. J.L. mainly took charge in the calculation part and focused on increasing the successful rate of the HT-calculations. F.X. developed the in-house code for VEC-compositions generation and HT calculations. L.Z. provided the suggestions for the manuscript. Y.Z. conceptualized, guided all the aspects and led the project as well as extensively revised the manuscript.

Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.actamat.2020.03.039.

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